- For the random walk, we saw that we could only return to state 0 in an even number of steps. (True for all states...)

- State 0 is said to have "<u>period 2</u>".

<u>Definition</u>: The <u>period</u> of a state i is the greatest common divisor of all integers $n \geq 1$ such that $p_{ii}^{(n)} > 0$.

<u>Claim</u>: If states i and j communicate, they have the same period.

("class property")

<u>Definition</u>: A Markov chain where all states have period 1 is an <u>aperiodic</u> chain.

<u>Theorem</u>: For an irreducible, positive recurrent, aperiodic Markov chain,

$$\lim_{n \to \infty} P_{ij}^{(n)}$$

exists and is independent of i.

Without "Irreducible"

Limit may exist but be dependent on starting state

$$\mathbb{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 1/2 & 1/2 & 0 & 0 \\ 3/4 & 1/4 & 0 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1/2 & 1/2 \end{bmatrix} \end{array}$$

Without "Irreducible"

Limit may exist but be dependent on starting state

$$\mathbb{P}^{(\infty)} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} 3/5 & 2/5 & 0 & 0 \\ 3/5 & 2/5 & 0 & 0 \\ 0 & 0 & 3/7 & 4/7 \\ 0 & 0 & 3/7 & 4/7 \end{bmatrix} \end{array}$$

<u>Theorem</u>: For an irreducible, positive recurrent, aperiodic Markov chain,

$$\lim_{n \to \infty} P_{ij}^{(n)}$$

exists and is independent of i.

⊛ An irreducible Markov chain on a finite state space is positive recurrent.

Thus, for an irreducible, aperiodic Markov chain on a finite state space

$$\lim_{n \to \infty} P_{ij}^{(n)}$$

exists and is independent of i.

<u>Definition</u>: An irreducible, positive recurrent aperiodic Markov chain is said to be <u>ergodic</u>.

"ergodic" = "well-behaved"

Consider an ergodic Markov chain $\{X_n\}$.

- Start at some state j.
- Suppose that you expect to return to j in 8 time steps.
- Once you return, by the Markov property, you again expect to return in 8 time steps.

- In the long-run, you will spend 1/8 of the time in state j.

- If you don't start at j, by irreducibility you will eventually hit j and then you will begin this "in the long-run" pattern.

Let $T_j = \min\{n \geq 1 : X_n = j\}$.

Then $\underbrace{E[T_j \mid X_0 = j]}_{E_j[T_j]}$ is the expected return time to j.

⊛ For an irreducible, recurrent, aperiodic Markov chain

$$\lim_{n\to\infty} P_{ij}^{(n)} = \frac{1}{E_j[T_j]}.$$